

Semester Test 3

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Electricity and Electronics EBN 111

7 June 2022

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	105 min	Open/Closed book:	Closed
Total number of pages (this page included)		20	

Lecturers: Prof X. Ye, Prof J.D. le Roux, Mr V. Dube**Assistant Lecturers:** N. Matic, K. Mphaga, J. Thiselton, D. Wepener, L. Zheve**ECSA GA assessment**

The following ECSA graduate attributes are assessed at entry level in this question paper: GA1 and GA2.

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Write your answers on the OCR forms provided. The question number **01** in this question paper corresponds to Row 01 of the OCR form, etc.
4. Read the instructions on the OCR form carefully.
5. Answers must include units where applicable.
6. Final answers must be written as real numbers and not as fractions. Use at least 4 significant figures to write numbers, including complex numbers.

INSTRUCTIONS**Do STEP 0 in the first 5 minutes of the session. (-5 to 0 min)****Do STEP 1 in the next 90 minutes of the test. (0-90 min)****Do STEP 2 within the next 15 minutes. (90-105 min)**

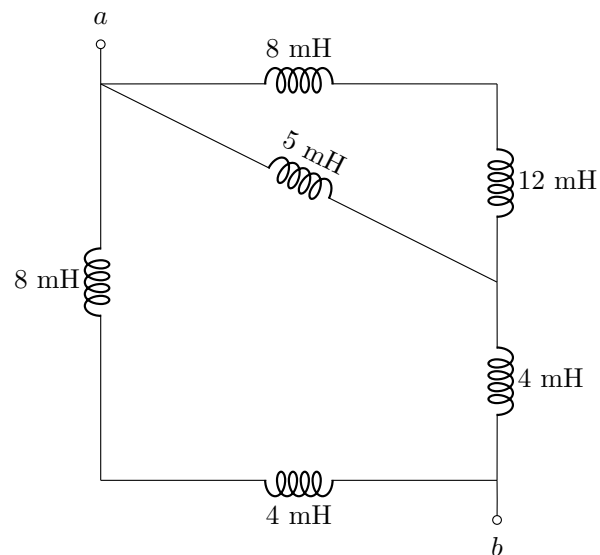
STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR forms.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 105 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

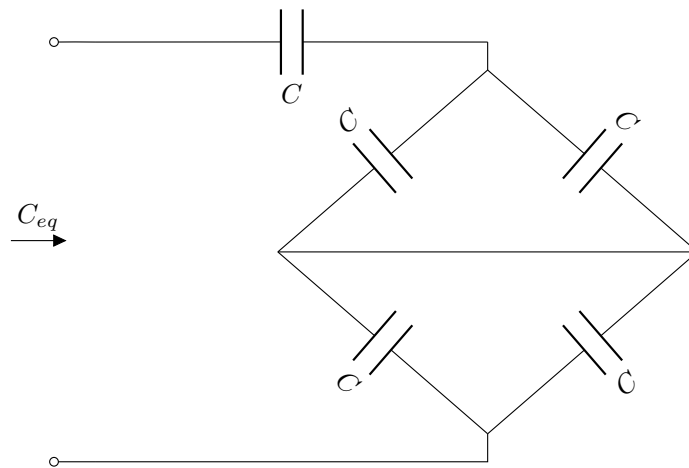
- The Assessment ID is: **2022EBN111S03**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Make use of full stops as decimal points.

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Section 1**Question 1 [2]**Find the equivalent inductance L_{ab} .**01** L_{ab} [2]

$$\begin{aligned} L_{ab} &= [(8+12) || (5+4)] || (8+4) \\ &= 8 || 12 \\ &= 4.8 \text{ mH} \end{aligned}$$

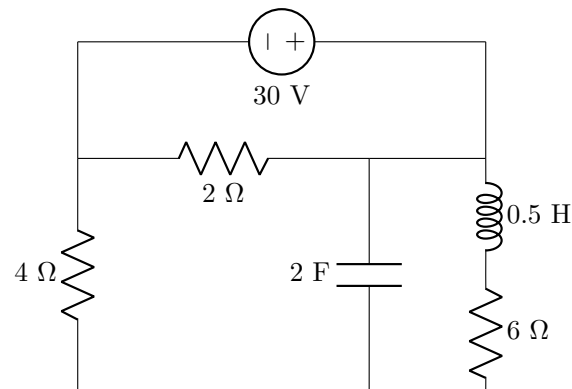
→

Question 2 [2]Find C_{eq} when $C = 2 \text{ mF}$.**02** C_{eq} [2]

$$\begin{aligned} C_{eq} &= C + 2C + 2C & \text{"+" : in series with} \\ &= 0.5C \\ &= 1 \text{ mF} \\ &\xrightarrow{\quad} D \end{aligned}$$

Questions 3 to 4 [4]

Find the energy stored in the capacitor ω_C , and the energy stored in the inductor ω_L under DC conditions.



03 ω_C [2]

04 ω_L [2]

The voltage source is in parallel with 4Ω .
 6Ω resistors.

$$i_L = \frac{30}{4+6} = 3 \Rightarrow \omega_L = \frac{1}{2} \times 0.5 \times 3^2 = 2.25 \text{ J}$$

$$V_C = 6 \times 3 = 18 \text{ V} \Rightarrow \omega_C = \frac{1}{2} \times 2 \times 18^2 = 324 \text{ J}$$

Question 5 [2]

The voltage across a 20 mH inductor is given by $v(t) = [4e^{-3t} + t]$ V. Determine the current $i(t)$ through the inductor at $t = 2$ s, assume that $i(0) = 1.2$ A.

05 $i(2 \text{ s})$ [2]

$$\begin{aligned}
 i(t) &= \frac{1}{L} \int_0^t v(t) dt + i(0) \\
 &= \frac{1}{0.02} \int_0^2 [4e^{-3t} + t] dt + 1.2 \\
 &= 50 \left(-\frac{4}{3} e^{-3t} + \frac{1}{2} t^2 \right)_0^2 + 1.2 \\
 &= 50 \left(-\frac{4}{3} e^{-6} + 2 + \frac{4}{3} \right) + 1.2 \\
 &= 50 \left(\frac{10}{3} - 3.305 \times 10^{-3} \right) + 1.2 \\
 &= \underline{167.701 \text{ A}}
 \end{aligned}$$

Total Section 1: [10]

Section 2

Questions 6 to 7 [2]

Given the sinusoidal signal:

$$v(t) = -50 \sin(20t + 20^\circ) \text{ V.}$$

06	Find the period of the sinusoid T . [1]
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07	Find $v(t)$ when $t = 10 \text{ ms}$. [1]
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06: $\omega = 20 \text{ rad/s}, \quad T = \frac{2\pi}{\omega} = 0.314 \text{ s.}$

07: $v(t) = -50 \sin(20 \times \overset{t}{\boxed{10 \times 10^{-3}}} + 20^\circ)$

$$v(10 \text{ ms}) = -50 \sin(\underbrace{0.549}_{\text{rad}})$$

$$\underline{v(10 \text{ ms}) = -26.092 \text{ V}} \rightarrow$$

Question 8 [2]

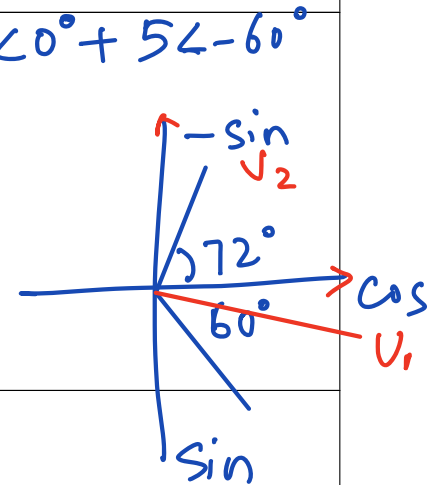
Simplify the expression for $\mathbf{Z}$.

$$\mathbf{Z} = \frac{2 + j3}{(1 - j3)^2} + \frac{6 - j8}{\sqrt{(-5 + j12)^*}}$$

08 $\mathbf{Z}$ [2]Hint: Y^* is the complex conjugate of Y .

$$\begin{aligned}
 \mathbf{Z} &= -\frac{17}{50} - j\frac{3}{25} + \frac{6 - j8}{\sqrt{-5 - j12}} \\
 &= -\frac{17}{50} - j\frac{3}{25} + \frac{6 - j8}{\sqrt{13} \angle -112.62^\circ} \\
 &= -\frac{17}{50} - j\frac{3}{25} + \frac{6 - j8}{\sqrt{13} \angle -56.31} \\
 &= 2.4292 + j0.0338 \\
 &= 2.4295 \angle 0.79832^\circ
 \end{aligned}$$

Questions 9 to 10 [4]	
Given two sinusoids: $v_1(t) = 12\cos(2t) + 5\sin(2t + 30^\circ) \text{ V, and}$ $v_2(t) = -15\sin(200t - 18^\circ) \text{ V}$	
09	Choose the correct statement. • Answer = 1 if $v_1(t)$ leads $v_2(t)$ by 30°. • Answer = 2 if $v_2(t)$ leads $v_1(t)$ by 30°. • Answer = 3 if $v_1(t)$ leads $v_2(t)$ by 60°. • Answer = 4 if $v_2(t)$ leads $v_1(t)$ by 60°. • Answer = 5 if none of the above statement is correct. Write 1, 2, 3, 4, or 5 in the answer block. [2]
10	$v_1(t)$ and $v_2(t)$ are represented by phasors $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively. Calculate $\mathbf{V}_T = \mathbf{V}_1 + \mathbf{V}_2$ [2]



$$v_1(t) = 15.13 \angle -16.627^\circ \text{ V}$$

[09]

V_2 leads V_1 by

$$72^\circ + 16.627^\circ = 88.627^\circ$$

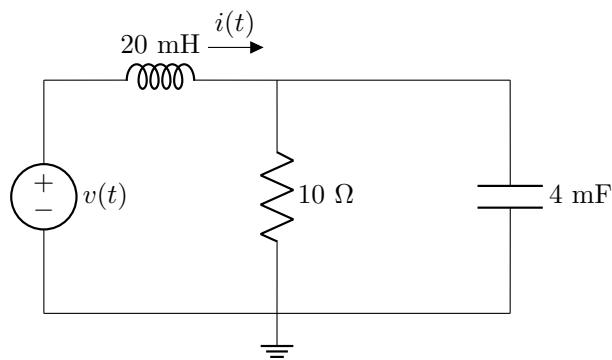
$$V_1 = 15.13 \angle -16.627^\circ \text{ V}$$

$$V_2 = 15 \angle 72^\circ \text{ V}$$

$$\Rightarrow V_T = 19.135 + j9.9358 \text{ V}$$

$$= 21.561 \angle 27.440^\circ \text{ V.}$$

Question 11 [2]

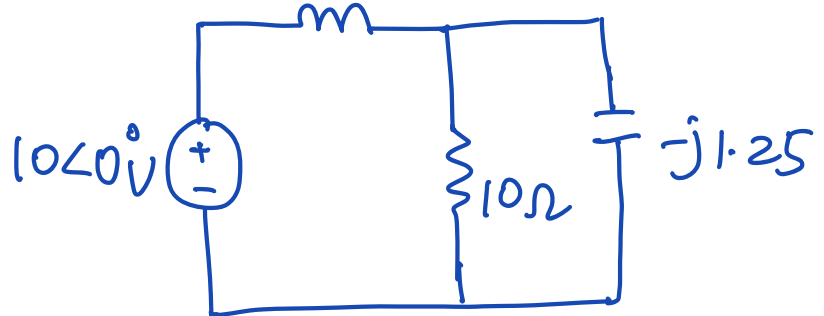
Find the current $i(t)$ at $t = 2$ s when $v(t) = 10 \cos(200t)$ V.11 $i(t=2 \text{ s})$ [2]

$$Z_L = j\omega L = j200 \times 20 \times 10^{-3} = j4 \Omega$$

$$Z_C = -j \frac{1}{\omega C}$$

$$= -j \frac{1}{200 \times 4 \times 10^{-3}}$$

$$= -j1.25$$



$$Z_T = j4 + \frac{10 \times (-j1.25)}{10 - j1.25}$$

$$= \frac{2}{13} + j\frac{36}{13} \Omega$$

$$\bar{I} = \frac{10 \angle 0^\circ}{Z_T} = 0.2 - j3.6 \text{ A}$$

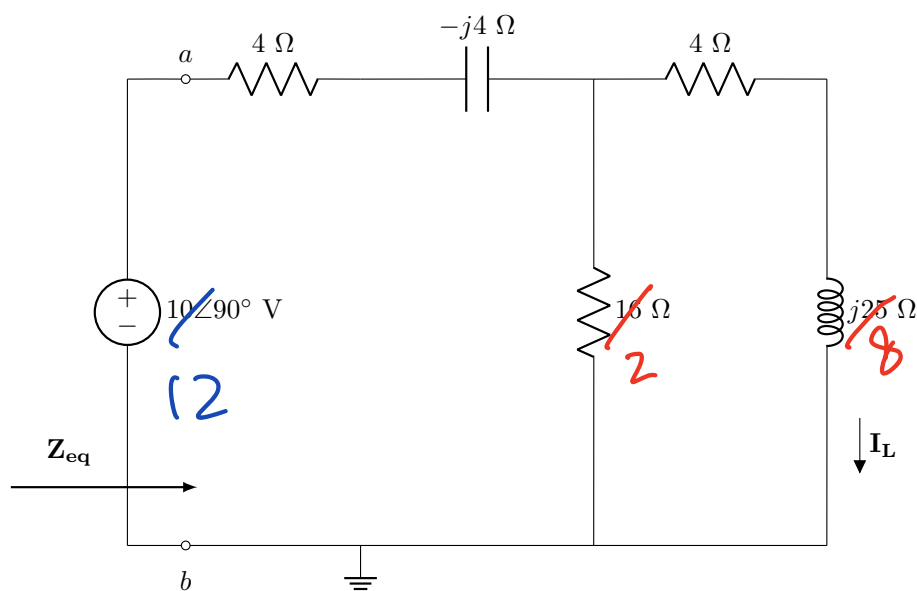
$$= 3.606 \angle -86.820^\circ \text{ A}$$

$$i(t) = 3.606 \cos(200t - 86.820^\circ)$$

$$\Rightarrow i(2) = 3.606 \cos\left(400 - \frac{86.820}{180} \cdot \pi\right)$$

$$= -3.169 \text{ A}$$

Questions 12 to 13 [4]

Find $\mathbf{Z}_{eq}$ with respect to terminals $a - b$, and the current through the inductor $\mathbf{I}_L$.12 $\mathbf{Z}_{eq}$ [2]13 $\mathbf{I}_L$ [2]

$$Z_1 = 4 - j4, \quad Z_2 = 4 + j8,$$

$$Z_3 = 2 \parallel Z_2 = \frac{2(4 + j8)}{6 + j8} = 1.76 + j0.32 \, \Omega.$$

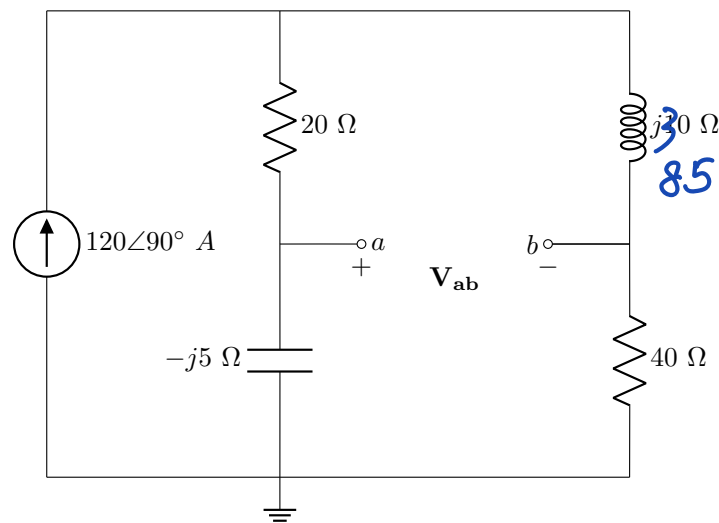
$$Z_{eq} = Z_1 + Z_3 = 5.76 - j3.68 \, \Omega.$$

$$I_L = j12 \cdot \frac{Z_3}{Z_1 + Z_3} \cdot \frac{1}{Z_2} = \frac{j12}{4 + j8} \cdot \frac{2(4 + j8)}{(6 + j8)(5.76 - j3.68)}$$

$$= 0.1233 + j0.3288 \, A$$

$$= 0.3511 \angle 69.44^\circ \, A.$$

Question 14 [2]

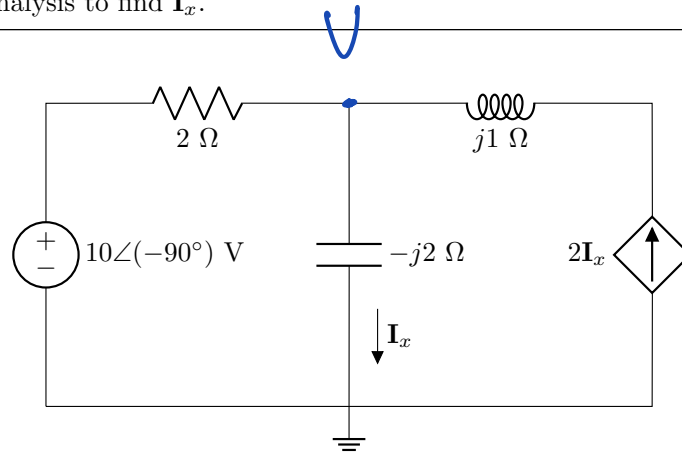
Find V_{ab} .14 V_{ab} [2]

$$\begin{aligned}
 Z_1 &= 20 - j5, \quad Z_2 = 40 + j85, \quad Z_1 + Z_2 = 60 + j80 \\
 V_a &= j120 \cdot \frac{Z_2}{Z_1 + Z_2} \cdot -j5 = 600 \cdot \frac{40 + j85}{60 + j80} \\
 V_b &= j120 \cdot \frac{Z_1}{Z_1 + Z_2} \cdot 40 = j4800 \cdot \frac{20 - j5}{60 + j80} \\
 V_{ab} &= V_a - V_b = \frac{600(40 + j85) - j4800(20 - j5)}{60 + j80} \\
 &= -360 - j270 \text{ V.}
 \end{aligned}$$

Total Section 2: [16]

Section 3

Question 15 [2]

Use nodal or mesh analysis to find $\mathbf{I}_x$.15 $\mathbf{I}_x$ [2]

$$\frac{V - (-j10)}{2} + \frac{V}{-j2} - 2 \cdot \frac{V}{-j2} = 0$$

$$\frac{V + j10}{2} = \frac{V}{-j2}$$

$$-j(V + j10) = V$$

$$\Rightarrow V + jV = 10 \Rightarrow V = \frac{10}{1+j} = 5 - j5 \text{ V}$$

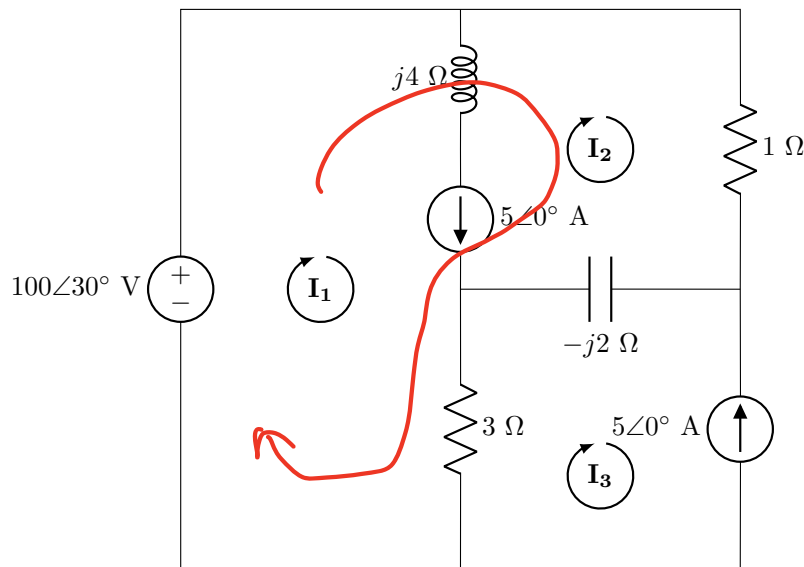
$$\mathbf{I}_x = \frac{V}{-j2} = \frac{5 - j5}{-j2} = \underline{2.5 + j2.5 \text{ A}}$$

Questions 16 to 19 [4]

Use mesh analysis to set up two equations in the form:

$$aI_1 + bI_2 = 100\angle 30^\circ - 15,$$

$$-j4I_1 + cI_2 = d.$$

Calculate unknown coefficients **a**, **b**, **c**, and **d**.

16 a [1]

17 b [1]

18 c [1]

19 d [1]

$$-100\angle 30^\circ + I_2 + (-j2)(I_2 - I_3) + 3(I_1 - I_3) = 0$$

$$-100\angle 30^\circ + (1 - j2)I_2 + j2I_3 + 3I_1 - 3I_3 = 0$$

$$3I_1 + (1 - j2)I_2 = (3 - j2)I_3 + 100\angle 30^\circ$$

$$= -15 + 50\sqrt{3} + j60$$

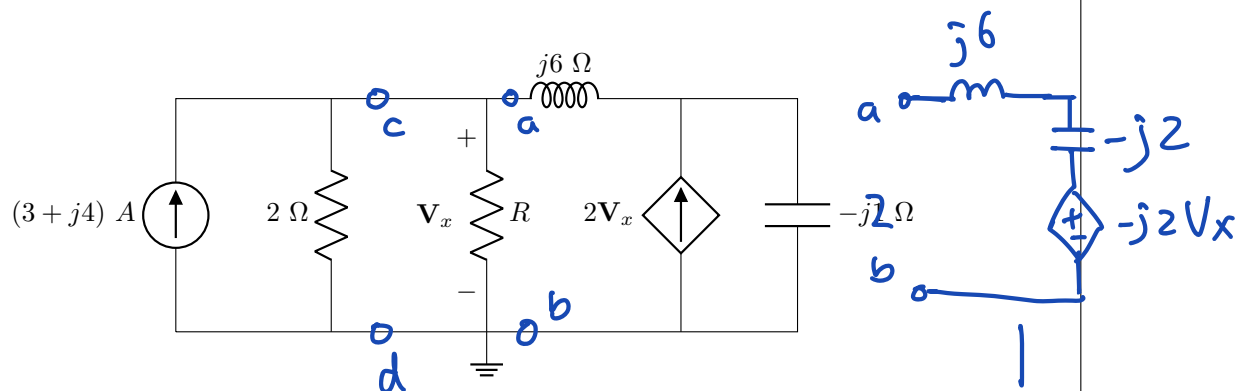
$$a = 1, b = -1$$

$$c = 1 - j2, d = 71.603 + j60$$

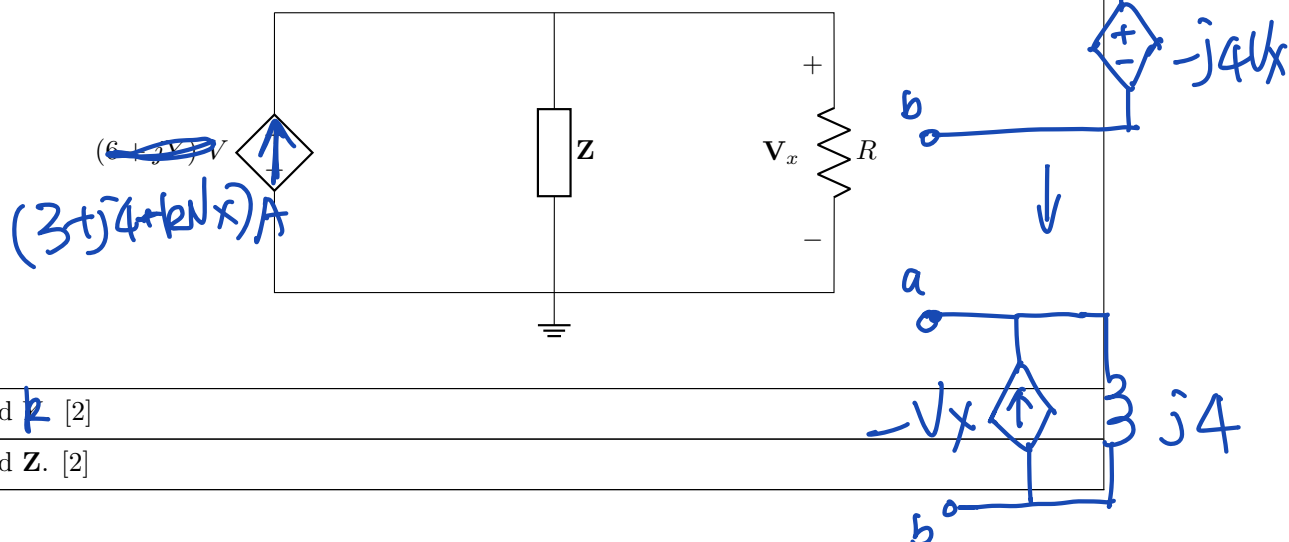
Questions 20 to 21 [4]

Use source transformation to transform Circuit A to Circuit B.

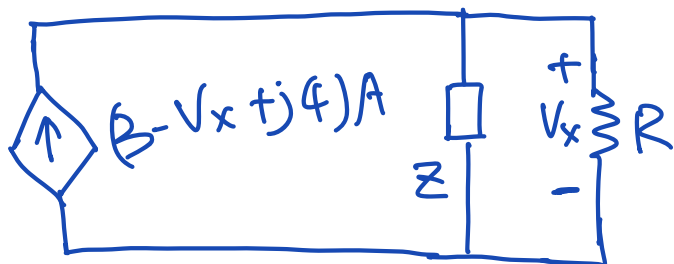
Circuit A



Circuit B

20 Find k [2]21 Find Z . [2]

when circuit area c-d link a-b, then

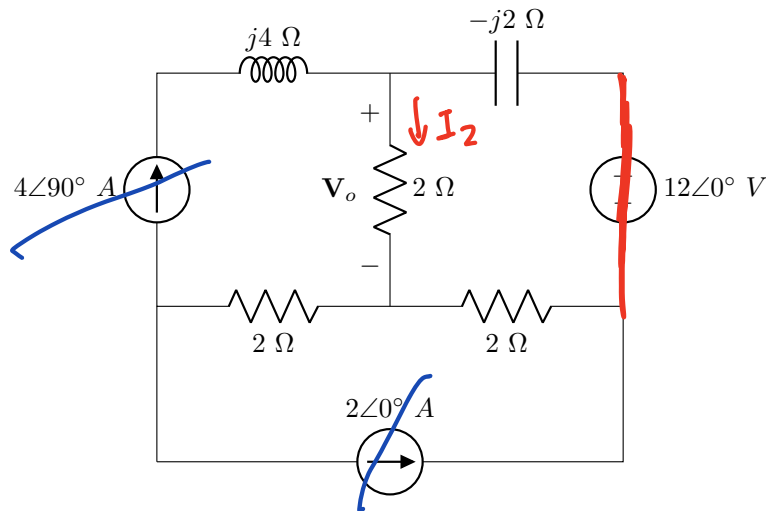


$$Z = \frac{2 \times j4}{2 + j4}$$

$$= 1.6 + j0.8 \Omega$$

$$k = -1$$

Questions 22 to 23 [4]



22 Find V_o^1 , the exclusive contribution from the independent voltage source. [2]

23 Find V_o^2 , the exclusive contribution from the current source $4\angle 90^\circ$ A. [2]

For V_o^1 , turn off two current sources, then they become open circuit

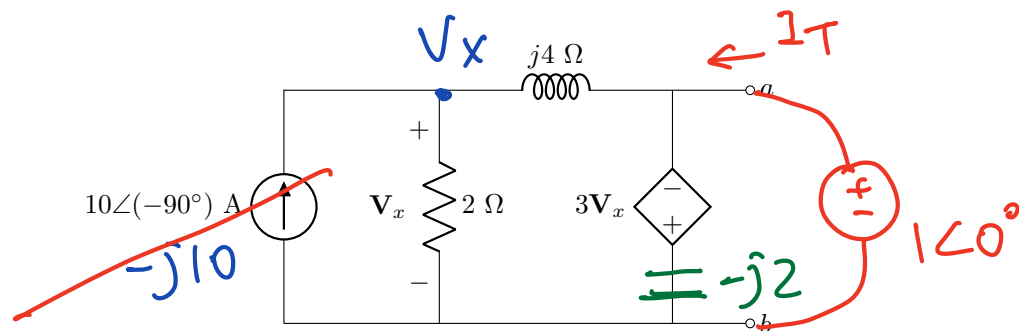
$$V_o^1 = 12 \cdot \frac{2}{4 - j2} = 4.8 + j2.4 \text{ V.}$$

For V_o^2 , the $2\angle 0^\circ$ A is open circuit, and voltage source is short circuit

$$I_2 = j4 \cdot \frac{2 - j2}{4 - j2}, \quad V_o = 2I_2 = j8 \cdot \frac{2 - j2}{4 - j2} = 1.6 + j4.8 \text{ V}$$

Questions 24 to 25 [4]

Determine the Thévenin equivalent voltage V_{Th} , and Thévenin equivalent impedance Z_{Th} with respect to terminals $a - b$.



24 V_{Th} [2]

25 Z_{Th} [2]

$$-j10 + \frac{V_x}{2} + \frac{V_x + 3V_x}{j4 - j2} = 0$$

$$\frac{V_x}{2} + \frac{4V_x}{j2} = j10, \quad (\times j2)$$

$$jV_x + 4V_x = -20 \Rightarrow V_x = \frac{-20}{4+j}$$

$$V_{Th} = -3V_x + (-j2) \cdot \frac{V_x + 3V_x}{j2} = -7V_x = \frac{-20}{4+j} = 32.941 - j8.235 \text{ V}$$

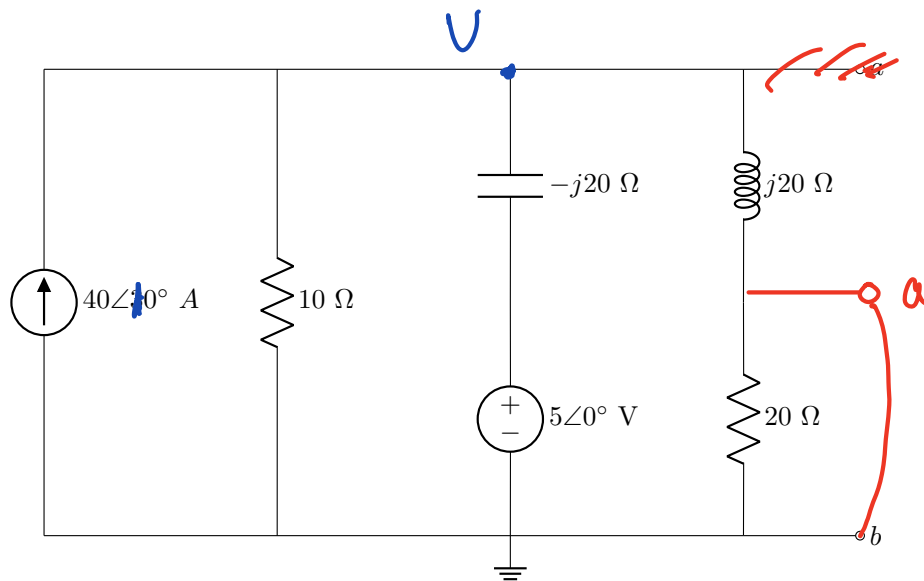
For Z_{Th} , add $V = 1\angle 0^\circ \text{ V}$, current source off.

$$V_x = 1 \cdot \frac{2}{2+j4} = \frac{2}{2+j4}, \quad I_T = \frac{1}{2+j4} + \frac{1+3V_x}{-j2}$$

$$I_T = \frac{1}{2+j4} + \frac{1}{-j2} - 3 \cdot \boxed{\frac{2}{2+j4}} \cdot \frac{1}{j2} = 0.7 + j0.6$$

$\downarrow$
 V_x

$$Z_{Th} = 0.8235 - j0.7059 \Omega$$

Question 26 [2]Determine the Norton equivalent current I_N with respect to terminals $a - b$.26 I_N [2]

Use nodal analysis

$$-40 + \frac{V}{10} + \frac{V-5}{-j20} + \frac{V}{j20} = 0 \quad \boxed{\times j20}$$

$$j2V - V + 5 + V = j800$$

$$j2V = -5 + j800$$

$$\Rightarrow V = \frac{-5 + j800}{j2}$$

$$I_N = \frac{V}{j20} = \frac{-5 + j800}{-40} = \underline{0.125 - j20 \text{ A}}$$

Total Section 3: [20]

Grand Total : [46]

PRACTICE ANSWER SHEET

Assessment ID:	2	0	2	2	E	B	N	1	1	1	S	0	3
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How to enter complex numbers:										
5	∠	-	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	-	<i>j</i>	8						
<i>j</i>	1	5							Ω	
-	1	.	5	+	<i>j</i>	8	.	2	m	V

Complex numbers can be entered in either **rectangular** or **polar** format.

Rectangular: The *j* is at the start of the imaginary number.

The *j* is in the CENTRE of the cell.

The *j* does NOT cross or touch cell boundaries!

Polar: The degree sign ° is directly after the angle value.

Please round irrational numbers to a minimum of 4 significant figures.

Answer		Unit
SECTION 1		
01	$L_a b =$	
02	$C_{eq} =$	
03	$\omega_C =$	
04	$\omega_L =$	
05	$i(2 \text{ ms}) =$	
SECTION 2		
06	$T =$	
07	$v(10 \text{ ms}) =$	
08	$\mathbf{Z} =$	
09	Answer =	No unit required
10	$\mathbf{V_T} =$	
11	$i(t = 2 \text{ s}) =$	
12	$\mathbf{Z_{eq}} =$	
13	$\mathbf{V_L} =$	
14	$\mathbf{V_{ab}} =$	
SECTION 3		
15	$\mathbf{I_x} =$	
16	$\mathbf{a} =$	No unit required
17	$\mathbf{b} =$	No unit required
18	$\mathbf{c} =$	No unit required
19	$\mathbf{d} =$	No unit required
20	$Y =$	No unit required
21	$\mathbf{Z} =$	
22	$\mathbf{V_o^1} =$	
23	$\mathbf{V_o^2} =$	
24	$\mathbf{V_{Th}} =$	
25	$\mathbf{Z_{Th}} =$	
26	$\mathbf{I_N} =$	